

Explainer: Problem 2 (Rudin 6.17) — Integration by Parts with a Monotone Weight

What's the claim?

For α monotonically increasing on $[a, b]$ and $g = G'$ continuous,

$$\int_a^b \alpha(x) g(x) dx = G(b)\alpha(b) - G(a)\alpha(a) - \int_a^b G d\alpha.$$

This is **integration by parts for Riemann–Stieltjes integrals**. The left side is an ordinary Riemann integral; the right side pairs α and G with the differentials swapped.

Why it's subtle

If α were also differentiable, this would just be $\int u dv = uv - \int v du$ using $d\alpha = \alpha' dx$. But here α is only assumed monotonic — no differentiability, possibly discontinuous (jumps). The proof has to work directly with partitions rather than invoking the standard product rule.

Key tool: the Mean Value Theorem

Given any partition $P = \{x_0, \dots, x_n\}$, on each subinterval (x_{i-1}, x_i) the MVT gives a point t_i with

$$g(t_i) \Delta x_i = G(x_i) - G(x_{i-1}).$$

This replaces a “Riemann sum with weight g ” by an exact telescoping difference in G .

Strategy

1. Pick a partition P and choose tags t_i via MVT as above.
2. Form the Riemann sum $\sum \alpha(x_i)g(t_i)\Delta x_i$ for $\int \alpha g dx$.
3. Substitute $g(t_i)\Delta x_i = G(x_i) - G(x_{i-1})$ and **apply Abel summation (summation by parts)**. Telescoping gives exactly

$$G(b)\alpha(b) - G(a)\alpha(a) - \sum_{i=1}^n G(x_{i-1})\Delta \alpha_i.$$

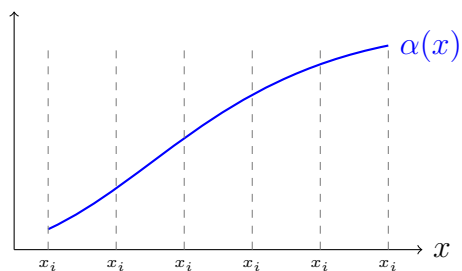
4. The last sum is a Riemann–Stieltjes sum for $\int G d\alpha$. Take $\text{mesh} \rightarrow 0$.

Abel summation reminder

$$\sum_{i=1}^n a_i(b_i - b_{i-1}) = a_n b_n - a_1 b_0 - \sum_{i=1}^{n-1} (a_{i+1} - a_i) b_i.$$

This is the discrete analogue of integration by parts and is the engine of the proof.

Picture



The weight α increases across the partition; the sum $\sum G(x_{i-1})\Delta\alpha_i$ weights G by the jumps of α .